

MANISH PATHAK

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Summary

High school student passionate about building and testing electronics. Hands-on experience with flight systems, microcontroller boards (ESP32, ESP8266, Raspberry Pi Pico), live telemetry tracking, and cybersecurity testing. I learn by building real projects, testing them, and improving iteratively.

Education

Kathmandu Model Secondary School (KMC)

Kathmandu, Nepal
High School Diploma – Science (Grade 11-12)

2082 – 2083 BS (Current)

Bhu.Pu Sainik Rising English Secondary School

Kathmandu, Nepal
Secondary Education Examination (SEE) – GPA: 3.85 / 4.0

Passed 2081 BS

Technical Skills

Electronics & Hardware

ESP32, ESP8266, Raspberry Pi Pico, NodeMCU, BMP180 pressure sensor, ADXL345 accelerometer, NRF24L01 wireless module, motor controllers, servo motors, telemetry systems, high-voltage circuits

Cybersecurity

Kali Linux, web penetration testing, network auditing, Burp Suite, Hydra, Hashcat, Wifite, Gobuster, Nmap, Metasploit, Wireshark

Programming Language

C Programming

Projects

Model Rocketry Telemetry & Avionics System

Personal Project

Status: On Hold

- Built a flight computer using ESP32 with BMP180 pressure sensor and ADXL345 accelerometer to track flight data
- Designed and tested experimental solid rocket motors using potassium nitrate and sugar; analyzed nozzle failures for design improvements
- Implemented high-speed telemetry transmission using NRF24L01 wireless modules and microSD card data logging

Wi-Fi Controlled Robotic Vehicle

Personal Project

Status: Completed

- Built robot chassis and power distribution using ESP32 and L298N motor drivers
- Developed custom Blynk smartphone application enabling wireless control with near-instant response time

SU-27 RC Aircraft – Smart Flight Control

Personal Project

Status: In Development

- Designing RC aircraft using Raspberry Pi Pico, electronic speed controllers, and brushless motors
- Replacing traditional radio remote with WebSocket-based Wi-Fi system for live telemetry and control mapping

Smart Assistive Glasses

Personal Project

Status: Completed

- Designed wearable frames with ESP8266 and dual ultrasonic distance sensors for object detection
- Programmed intelligent buzzer logic: frequency proportional to object proximity, enabling safe navigation for visually impaired users

Home Appliance Relay Controller

Personal Project

Status: Completed

- Engineered isolated 2-channel relay panel for safe high-voltage appliance circuit control
- Integrated ESP8266 with Blynk platform for secure remote appliance management via smartphone

Mini Solid-State Tesla Coil

Personal Project

Status: Completed

- Constructed primary and secondary induction coils tuned for resonant high-frequency output
- Built driver circuit with 2N2222A transistor and 9V supply to wirelessly energize gas-discharge lamps

Cybersecurity Practice

TryHackMe – Online Security Labs

Self-Directed Practice

Status: Active

- Completed 48 security challenge rooms and earned 7 specialization badges (Global Rank: 311,928)
- Practiced password cracking (Hashcat, John the Ripper), automated reconnaissance (Gobuster, Wifite, Hydra), web application analysis (Burp Suite), and network scanning (Nmap)
- Configured and maintained live-boot Kali Linux environment for secure, isolated lab operations